

# ■■ Cybersecurity Notes

## Day 1 – Step 4 : Windows Permissions

Privilege Escalation • Users & Groups • Security Tab

### ■ Permission Kya Hoti Hai?

Windows mein har ek file aur folder pe ek **access control** hota hai.

Yeh decide karta hai ki:

- Kaun us file ko **open (Read)** kar sakta hai
- Kaun us file ko **badal (Write/Edit)** sakta hai
- Kaun us file ko **delete ya control (Full Control)** kar sakta hai

Isi cheez ko kehte hain ■ **Permissions**

### ■ 3 Main Permission Types — Ek Nazar Mein

| Permission            | Matlab / Meaning                                                         | Example                                                                    |
|-----------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------|
| ■ <b>Read</b>         | Sirf dekh sakte ho, kuch badal nahi sakte                                | Ek student ke paas class notes ki Read permission hoti hai                 |
| —■ <b>Write</b>       | File mein kuch likhna, edit karna ya naya file banana                    | Teacher notes mein changes kar sakta hai                                   |
| ■ <b>Full Control</b> | Sab kuch kar sakte ho — Read, Write, Delete, Permission bhi change karo! | Principal sab kuch kar sakta hai — notes likho, mitao, access bhi do ya lo |

### ■ Users & Groups — Security Tab Mein Kya Dikhta Hai?

Jab tum kisi folder ki **Properties** → **Security tab** kholte ho, yeh teen main entries dikhti hain:

| User / Group     | Kaun Hai Yeh?                                    | Default Permission  | Cyber Risk                           |
|------------------|--------------------------------------------------|---------------------|--------------------------------------|
| ■■ <b>SYSTEM</b> | Windows OS khud — background mein kaam karta hai | <b>Full Control</b> | Agar hack ho toh poora OS vulnerable |

|                  |                                           |              |                                |
|------------------|-------------------------------------------|--------------|--------------------------------|
| ■ Administrators | Admin users — powerful accounts hote hain | Full Control | Hackers isko target karte hain |
| ■ Users          | Normal users — limited powers hoti hain   | Read / Write | Sabse safe — limited access    |

■ Aasaan Bhasha Mein Samjho — School Example:

SYSTEM = School ki building (sabse zyada power, background mein kaam karta hai)

Administrators = Principal (poori school control kar sakta hai)

Users = Students (sirf apna kaam kar sakte hain, baaki nahi)

■ Privilege Escalation — Hacker Ka Sabse Bada Weapon

■■ Privilege Escalation Kya Hoti Hai?

Jab koi hacker ya malware apne aap ko **zyada permissions de leta hai** — jitni usse milni chahiye thi usse zyada.

Simple Example:

Tum school mein **student** ho → sirf apna kaam kar sakte ho

Hacker exploit karke **Principal ban jaata hai** → ab poori school control mein!

**Yahi hota hai jab hacker User → Administrator → SYSTEM level tak pahunch jaata hai.**

Privilege Escalation ke 3 Steps:

| Step         | Hacker Kya Karta Hai?                                                    | Real World Example                           |
|--------------|--------------------------------------------------------------------------|----------------------------------------------|
| 1■■ Entry    | Kisi normal user account mein ghusta hai — password crack ya phishing se | Tumhara email hack karke login kar leta hai  |
| 2■■ Escalate | Vulnerability use karke apni permission badhata hai — User se Admin      | System ki ek purani file ka fayda uthata hai |
| 3■■ Control  | Ab Full Control mil gayi — virus install, data chori, sabhi files access | Ab poora system uske control mein hai        |

■ Practical Task — ProgramData Ka Permission Check Karo

■■ Yeh steps follow karo:

1. Windows key + E dabao → File Explorer khulega
2. Address bar mein likho: C:\ProgramData → Enter dabao
3. Kisi bhi folder pe Right Click karo
4. "Properties" pe click karo
5. Upar "Security" tab pe click karo

**Ab observe karo aur note karo:**

- Kaun kaun se Users / Groups dikhe? (SYSTEM, Administrators, Users?)
- Kisko Full Control mila hua hai?
- Kisi normal User ko Full Control hai kya? (Agar hai toh — RED FLAG! ■■)

#### ■ Kya Normal Hota Hai vs Kya Suspicious Hai?

- NORMAL — SYSTEM aur Administrators ko Full Control (yeh expected hai)
- NORMAL — Users ko sirf Read / Write milti hai
- SUSPICIOUS — Unknown user ko Full Control mila hua hai
- SUSPICIOUS — Normal user account ko bhi Full Control hai
- SUSPICIOUS — Koi naya user dikh raha hai jo tumne create nahi kiya

### ■ Complete Permission Chart — Ek Hi Jagah Sab Kuch

| Permission     | Read | Write | Execute | Delete | Change Permission |
|----------------|------|-------|---------|--------|-------------------|
| ■ Read Only    | ■    | ■     | ■       | ■      | ■                 |
| —■ Read+Write  | ■    | ■     | ■       | ■      | ■                 |
| ■■ Modify      | ■    | ■     | ■       | ■      | ■                 |
| ■ Full Control | ■    | ■     | ■       | ■      | ■                 |

### ■ Quick Revision — Ek Nazar Mein Sab Kuch

|                                                                                           |                                                                                                    |
|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <p>■ <b>Permission</b></p> <p>File/folder ka access control — kaun kya kar sakta hai</p>  | <p>■ <b>Read</b></p> <p>Sirf dekho — edit ya delete nahi kar sakte</p>                             |
| <p>—■ <b>Write</b></p> <p>Badal sakte ho — naya likh ya edit kar sakte ho</p>             | <p>■ <b>Full Control</b></p> <p>Sab kuch — Read, Write, Delete, aur permission bhi change karo</p> |
| <p>■■ <b>SYSTEM</b></p> <p>OS khud — sabse zyada powerful, background mein chalta hai</p> | <p>■ <b>Administrators</b></p> <p>Admin users — Full Control, hackers ka pehla target</p>          |
| <p>■ <b>Users</b></p> <p>Normal users — Limited access, sabse safe</p>                    | <p>■ <b>Privilege Escal.</b></p> <p>Hacker User se Admin ban jaata hai — MOST DANGEROUS ATTACK</p> |

■ Day 1 – Step 4 Complete! Privilege Escalation Samajh Aa Gayi? Keep Going! ■